

Integration:

“The inverse process of differentiation i.e the process of finding such a function whose derivative is given is called anti-derivation or integration”.

For example:

An anti-derivative of $2x$ is x^2 since $\frac{d}{dx}(x^2) = 2x$.

Indefinite Integral:

$\int f(x)dx = F(x) + c$ is called the indefinite integral of $f(x)$, where $f(x)$ is integrand, $F(x)$ is particular integral and c is the constant of integration.

Theorems on Integration:

(i) **Prove that** $\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c, n \neq -1$

Proof:

$$\text{Since } \frac{d}{dx}[f(x)]^{n+1} = (n+1)[f(x)]^n f'(x)$$

So by definition

$$\int (n+1)[f(x)]^n f'(x)dx = [f(x)]^{n+1} + c_1$$

$$(n+1) \int [f(x)]^n f'(x)dx = [f(x)]^{n+1} + c_1$$

$$\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + \frac{c_1}{n+1}$$

$$\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c \quad \because c = \frac{c_1}{n+1}$$

(ii) **Prove that**

$$\int [f(x)]^{-1} f'(x)dx = \ln f(x) + c, f(x) > 0$$

Proof:

$$\text{Since } \frac{d}{dx} \ln[f(x)] = \frac{1}{f(x)} f'(x)$$

By definition we have

$$\int \frac{1}{f(x)} f'(x)dx = \ln f(x) + c, f(x) > 0$$

Standard Formulae for Anti-Derivatives:

General Form	Simple Form
$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c, (n \neq -1)$	$\int x^n dx = \frac{x^{n+1}}{n+1} + c, (n \neq -1)$
$\int \sin(ax+b) dx = \frac{-\cos(ax+b)}{a} + c$	$\int \sin x dx = -\cos x + c$
$\int \cos(ax+b) dx = \frac{\sin(ax+b)}{a} + c$	$\int \cos x dx = \sin x + c$
$\int \sec^2(ax+b) dx = \frac{\tan(ax+b)}{a} + c$	$\int \sec^2 x dx = \tan x + c$
$\int \operatorname{cosec}^2(ax+b) dx = \frac{-\cot(ax+b)}{a} + c$	$\int \operatorname{cosec}^2 x dx = -\cot x + c$
$\int \sec(ax+b) \tan(ax+b) dx = \frac{\sec(ax+b)}{a} + c$	$\int \sec x \tan x dx = \sec x + c$
$\int \operatorname{cosec}(ax+b) \cot(ax+b) dx = \frac{-\operatorname{cosec}(ax+b)}{a} + c$	$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$
$\int e^{\lambda x + \mu} dx = \frac{e^{\lambda x + \mu}}{\lambda} + c, (\lambda \neq 0)$	$\int e^x dx = e^x + c$
$\int a^{\lambda x + \mu} dx = \frac{a^{\lambda x + \mu}}{\lambda \ln a} + c, (a > 0, a \neq 1, \lambda \neq 0)$	$\int a^x dx = \frac{a^x}{\ln a} + c (a > 0, a \neq 1)$
$\int \frac{1}{ax+b} dx = \frac{\ln ax+b }{a} + c, (ax+b \neq 0)$	$\int \frac{1}{x} dx = \ln x + c, x \neq 0$
$\int \tan(ax+b) dx = \frac{1}{a} \ln \sec(ax+b) + c$ $= -\frac{1}{a} \ln \cos(ax+b) + c$	$\int \tan x dx = \ln \sec x + c$ $= -\ln \cos x + c$
$\int \cot(ax+b) dx = \frac{1}{a} \ln \sin(ax+b) + c$	$\int \cot x dx = \ln \sin x + c$
$\int \sec(ax+b) dx = \frac{1}{a} \ln \sec(ax+b) + \tan(ax+b) + c$	$\int \sec x dx = \ln \sec x + \tan x + c$
$\int \operatorname{cosec}(ax+b) dx = \frac{1}{a} \ln \operatorname{cosec}(ax+b) - \cot(ax+b) + c$	$\int \operatorname{cosec} x dx = \ln \operatorname{cosec} x - \cot x + c$

EXERCISE 3.2**Q.1 Evaluate the following indefinite integrals**

(i) $\int (3x^2 - 2x + 1)dx$

Solution:

$$\begin{aligned} & \int (3x^2 - 2x + 1)dx \\ &= \int 3x^2 dx - \int 2x dx + \int 1 dx \\ &= 3 \int x^2 dx - 2 \int x dx + 1 \int dx \end{aligned}$$

$$\text{Using } \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\begin{aligned} &= 3 \left(\frac{x^3}{3} \right) - 2 \left(\frac{x^2}{2} \right) + x + c \\ &= x^3 - x^2 + x + c \end{aligned}$$

(ii) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx, (x > 0)$

Solution:

$$\begin{aligned} & \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx \\ &= \int x^{\frac{1}{2}} dx + \int x^{-\frac{1}{2}} dx \end{aligned}$$

$$= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + c$$

$$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c$$

(iii) $\int x(\sqrt{x} + 1)dx, (x > 0)$

Solution:

$$\begin{aligned} & \int x(\sqrt{x} + 1)dx \\ &= \int x\sqrt{x}dx + \int x dx \\ &= \int x^{\frac{3}{2}}dx + \int x dx \end{aligned}$$

$$= \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} + \frac{x^2}{2} + c$$

$$= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{x^2}{2} + c$$

$$= \frac{2}{5} x^{\frac{5}{2}} + \frac{1}{2} x^2 + c$$

(iv) $\int (2x+3)^{\frac{1}{2}} dx$

Solution:

$$\int (2x+3)^{\frac{1}{2}} dx$$

Multiplying and dividing by 2

$$= \frac{1}{2} \int (2x+3)^{\frac{1}{2}} (2) dx$$

$$= \frac{1}{2} \frac{(2x+3)^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c$$

$$= \frac{1}{2} \frac{(2x+3)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + c$$

$$= \frac{1}{2} \times \frac{2}{3} (2x+3)^{\frac{3}{2}} + c$$

$$= \frac{1}{3} (2x+3)^{\frac{3}{2}} + c$$

(v) $\int (\sqrt{x} + 1)^2 dx, (x > 0)$

Solution:

$$\int (\sqrt{x} + 1)^2 dx$$

$$= \int (x + 2\sqrt{x} + 1) dx$$

$$= \int x dx + 2 \int x^{\frac{1}{2}} dx + \int 1 dx$$

$$= \frac{x^2}{2} + \frac{2x^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + x + c$$

$$= \frac{x^2}{2} + \frac{4}{3} x^{\frac{3}{2}} + x + c$$

(vi) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 dx, (x > 0)$

Solution:

$$\begin{aligned} & \int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int \left(x + \frac{1}{x} - 2 \right) dx \\ &= \int x dx + \int x^{-1} dx - 2 \int dx \\ &= \frac{x^2}{2} + \ln|x| - 2x + c \\ &\because \int \frac{1}{x} dx = \ln|x| + c \end{aligned}$$

(vii) $\int \frac{3x+2}{\sqrt{x}} dx, (x > 0)$

Solution:

$$\begin{aligned} & \int \frac{3x+2}{\sqrt{x}} dx \\ &= \int \left(\frac{3x}{\sqrt{x}} + \frac{2}{\sqrt{x}} \right) dx \\ &= \int 3x^{\frac{1}{2}} dx + \int 2x^{\frac{-1}{2}} dx \\ &= 3 \int x^{\frac{1}{2}} dx + 2 \int x^{\frac{-1}{2}} dx \\ &= 3 \frac{x^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} + 2 \frac{x^{\frac{-1}{2}+1}}{\left(\frac{-1}{2}+1\right)} + c \\ &= 3 \frac{x^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + 2 \frac{x^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c \\ &= 2x^{\frac{3}{2}} + 4x^{\frac{1}{2}} + c \end{aligned}$$

(viii) $\int \frac{\sqrt{y}(y+1)}{y} dy, (y > 0)$

Solution:

$$\int \frac{\sqrt{y}(y+1)}{y} dy$$

$$\begin{aligned} &= \int \left(\frac{y+1}{\sqrt{y}} \right) dy \\ &= \int \frac{y}{\sqrt{y}} dy + \int \frac{1}{\sqrt{y}} dy \\ &= \int y^{\frac{1}{2}} dy + \int y^{\frac{-1}{2}} dy \\ &= \frac{y^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} + \frac{y^{\frac{-1}{2}+1}}{\left(\frac{-1}{2}+1\right)} + c \\ &= \frac{y^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + \frac{y^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c \\ &= \frac{2}{3} y^{\frac{3}{2}} + 2y^{\frac{1}{2}} + c \end{aligned}$$

(ix) $\int \frac{(\sqrt{\theta}-1)^2}{\sqrt{\theta}} d\theta, (\theta > 0)$

Solution:

$$\begin{aligned} & \int \frac{(\sqrt{\theta}-1)^2}{\sqrt{\theta}} d\theta \\ &= \int \frac{(\theta+1-2\sqrt{\theta})}{\sqrt{\theta}} d\theta \\ &= \int \left(\frac{\theta}{\sqrt{\theta}} + \frac{1}{\sqrt{\theta}} - \frac{2\sqrt{\theta}}{\sqrt{\theta}} \right) d\theta \\ &= \int \theta^{\frac{1}{2}} d\theta + \int \theta^{\frac{-1}{2}} d\theta - 2 \int d\theta \\ &= \frac{\theta^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} + \frac{\theta^{\frac{-1}{2}+1}}{\left(\frac{-1}{2}+1\right)} - 2\theta + c \\ &= \frac{\theta^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + \frac{\theta^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} - 2\theta + c \end{aligned}$$

$$= \frac{2}{3} \theta^{\frac{3}{2}} + 2\theta^{\frac{1}{2}} - 2\theta + c$$

$$(x) \quad \int \frac{(1-\sqrt{x})^2}{\sqrt{x}} dx, (x > 0)$$

Solution:

$$\begin{aligned} & \int \frac{(1-\sqrt{x})^2}{\sqrt{x}} dx \\ &= \int \frac{(1+x-2\sqrt{x})}{\sqrt{x}} dx \\ &= \int \left(\frac{1}{\sqrt{x}} + \frac{x}{\sqrt{x}} - \frac{2\sqrt{x}}{\sqrt{x}} \right) dx \\ &= \int x^{-\frac{1}{2}} dx + \int x^{\frac{1}{2}} dx - 2 \int dx \\ &= \frac{x^{-\frac{1}{2}+1}}{\left(-\frac{1}{2}+1\right)} + \frac{x^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} - 2x + c \\ &= \frac{x^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + \frac{x^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} - 2x + c \\ &= 2x^{\frac{1}{2}} + \frac{2}{3}x^{\frac{3}{2}} - 2x + c \end{aligned}$$

$$(xi) \quad \int \frac{e^{2x} + e^x}{e^x} dx$$

Solution:

$$\begin{aligned} & \int \frac{e^{2x} + e^x}{e^x} dx \\ &= \int \frac{e^{2x}}{e^x} dx + \int \frac{e^x}{e^x} dx \\ &= \int e^x dx + \int dx \\ &= e^x + x + c \end{aligned}$$

Q.2 Evaluate

$$(i) \quad \int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}}, \begin{pmatrix} x+a > 0 \\ x+b > 0 \end{pmatrix}$$

Solution:

$$\int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}}$$

By rationalizing the denominator

$$\begin{aligned} &= \int \frac{1}{\sqrt{x+a} + \sqrt{x+b}} \times \frac{\sqrt{x+a} - \sqrt{x+b}}{\sqrt{x+a} - \sqrt{x+b}} dx \\ &= \int \frac{\sqrt{x+a} - \sqrt{x+b}}{(\sqrt{x+a})^2 - (\sqrt{x+b})^2} dx \\ &= \int \frac{\sqrt{x+a} - \sqrt{x+b}}{x+a-x-b} dx \\ &= \frac{1}{a-b} \left(\int (x+a)^{\frac{1}{2}} dx - \int (x+b)^{\frac{1}{2}} dx \right) \\ &= \frac{1}{a-b} \left(\frac{(x+a)^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} - \frac{(x+b)^{\frac{1}{2}+1}}{\left(\frac{1}{2}+1\right)} \right) + c \\ &= \frac{1}{(a-b)} \left(\frac{2}{3}(x+a)^{\frac{3}{2}} - \frac{2}{3}(x+b)^{\frac{3}{2}} \right) + c \\ &= \frac{2}{3(a-b)} \left((x+a)^{\frac{3}{2}} - (x+b)^{\frac{3}{2}} \right) + c \end{aligned}$$

$$(ii) \quad \int \frac{1-x^2}{1+x^2} dx$$

Solution:

$\frac{1-x^2}{1+x^2}$ is an improper fraction therefore

$$x^2 + 1 \sqrt{\frac{-1}{-x^2+1}} = \frac{\sqrt{-x^2+1}}{2}$$

$$\text{So, } \int \frac{1-x^2}{1+x^2} dx$$

$$= \int \left(-1 + \frac{2}{1+x^2} \right) dx$$

$$= -\int 1 dx + 2 \int \frac{1}{1+x^2} dx$$

$$= -x + 2 \tan^{-1} x + c \because \int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$= 2 \tan^{-1} x - x + c$$

$$(iii) \int \frac{dx}{\sqrt{x+a} + \sqrt{x}}, (x > 0, a > 0)$$

Solution:

$$\int \frac{dx}{\sqrt{x+a} + \sqrt{x}}$$

By rationalizing the denominator

$$= \int \frac{1}{\sqrt{x+a} + \sqrt{x}} \times \frac{\sqrt{x+a} - \sqrt{x}}{\sqrt{x+a} - \sqrt{x}} dx$$

$$= \int \frac{\sqrt{x+a} - \sqrt{x}}{(\sqrt{x+a})^2 - (\sqrt{x})^2} dx$$

$$= \int \frac{\sqrt{x+a} - \sqrt{x}}{x+a-x} dx$$

$$= \frac{1}{a} \left(\int (x+a)^{\frac{1}{2}} dx - \int x^{\frac{1}{2}} dx \right)$$

$$= \frac{1}{a} \left(\frac{(x+a)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} - \frac{x^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} \right) + c$$

$$= \frac{1}{a} \left(\frac{2}{3} (x+a)^{\frac{3}{2}} - \frac{2}{3} x^{\frac{3}{2}} \right) + c$$

$$= \frac{2}{3a} \left((x+a)^{\frac{3}{2}} - x^{\frac{3}{2}} \right) + c$$

$$(iv) \int (a-2x)^{\frac{3}{2}} dx$$

Solution:

$$\int (a-2x)^{\frac{3}{2}} dx$$

$$= -\frac{1}{2} \int (a-2x)^{\frac{3}{2}} (-2) dx$$

$$= -\frac{1}{2} \left(\frac{(a-2x)^{\frac{3}{2}+1}}{\left(\frac{3}{2}+1\right)} \right) + c$$

$$= -\frac{1}{2} \left(\frac{(a-2x)^{\frac{5}{2}}}{\left(\frac{5}{2}\right)} \right) + c$$

$$= -\frac{1}{5} (a-2x)^{\frac{5}{2}} + c$$

$$(v) \int \frac{(1+e^x)^3}{e^x} dx$$

Solution:

$$\int \frac{(1+e^x)^3}{e^x} dx$$

Using $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

$$= \int \left(\frac{(1+e^{3x} + 3(1)(e^x)(1+e^x))}{e^x} \right) dx$$

$$= \int \left(\frac{(1+e^{3x} + 3e^x + 3e^{2x})}{e^x} \right) dx$$

$$= \int \left(\frac{1}{e^x} + \frac{e^{3x}}{e^x} + \frac{3e^x}{e^x} + \frac{3e^{2x}}{e^x} \right) dx$$

$$= \int e^{-x} dx + \int e^{2x} dx + 3 \int dx + 3 \int e^x dx$$

$$= \frac{e^{-x}}{-1} + \frac{e^{2x}}{2} + 3x + 3e^x + c$$

$$= -\frac{1}{e^x} + \frac{e^{2x}}{2} + 3x + 3e^x + c$$

$$(vi) \int \sin(a+b)x dx$$

Solution:

$$\int \sin(a+b)x dx$$

$$= \frac{1}{a+b} (-\cos(a+b)x) + c$$

$$= -\frac{1}{(a+b)} \cos(a+b)x + c$$

$$(vii) \int \sqrt{1-\cos 2x} dx, (1-\cos 2x > 0)$$

Solution:

$$\int \sqrt{1-\cos 2x} dx$$

$$= \int \sqrt{2\sin^2 x} dx \because 1-\cos 2x = 2\sin^2 x$$

$$= \sqrt{2} \int \sin x dx$$

$$= \sqrt{2}(-\cos x) + c$$

(viii) $\int (\ln x) \times \frac{1}{x} dx, (x > 0)$

Solution:

$$\int (\ln x) \times \frac{1}{x} dx$$

$$= \frac{(\ln x)^2}{2} + c$$

(ix) $\int \sin^2 x dx$

Solution:

$$\int \sin^2 x dx$$

$$\because \sin^2 x = \left(\frac{1 - \cos 2x}{2} \right)$$

$$= \int \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= \frac{1}{2} \left[\int (1 - \cos 2x) dx \right]$$

$$= \frac{1}{2} \left[\int dx - \int \cos 2x dx \right]$$

$$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + c$$

$$= \frac{1}{2} x - \frac{1}{4} \sin 2x + c$$

(x) $\int \frac{1}{1 + \cos x} dx, \left(-\frac{\pi}{2} < x < \frac{\pi}{2} \right)$

Solution:

$$\int \frac{1}{1 + \cos x} dx$$

$$= \int \frac{1}{2 \cos^2 \frac{x}{2}} dx \because 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$= \frac{1}{2} \left(\int \sec^2 \frac{x}{2} \right) dx$$

$$= -\sqrt{2} \cos x + c$$

$$= \frac{1}{2} \left(\frac{\tan \frac{x}{2}}{\left(\frac{1}{2} \right)} \right) + c$$

$$= \tan \frac{x}{2} + c$$

(xi) $\int \frac{ax + b}{ax^2 + 2bx + c} dx$

Solution:

$$\int \frac{ax + b}{ax^2 + 2bx + c} dx$$

$$= \frac{1}{2} \int \left(\frac{2ax + 2b}{ax^2 + 2bx + c} \right) dx$$

$$= \frac{1}{2} \ln |ax^2 + 2bx + c| + c_1$$

(xii) $\int \cos 3x \sin 2x dx$

Solution:

$$\int \cos 3x \sin 2x dx$$

$$= \frac{1}{2} \int (2 \cos 3x \sin 2x) dx$$

Using

$$2 \cos \alpha \sin \alpha = \sin(\alpha + \beta) - \sin(\alpha - \beta)$$

$$= \frac{1}{2} \int [\sin(3x + 2x) - \sin(3x - 2x)] dx$$

$$= \frac{1}{2} \left(\int \sin 5x dx - \int \sin x dx \right)$$

$$= \frac{1}{2} \left[\frac{-\cos 5x}{5} - (-\cos x) \right] + c$$

$$= -\frac{1}{2} \left(\frac{\cos 5x}{5} - \cos x \right) + c$$

(xiii) $\int \frac{\cos 2x - 1}{1 + \cos 2x} dx, (1 + \cos 2x \neq 0)$

Solution:

$$\begin{aligned} & \int \frac{\cos 2x - 1}{1 + \cos 2x} dx \\ &= -\int \left(\frac{1 - \cos 2x}{1 + \cos 2x} \right) dx \end{aligned}$$

As

$$1 + \cos 2x = 2 \cos^2 x$$

$$1 - \cos 2x = 2 \sin^2 x$$

$$\begin{aligned} &= -\int \frac{2 \sin^2 x}{2 \cos^2 x} dx \\ &= -\int \tan^2 x dx \\ &= -\int (\sec^2 x - 1) dx \\ &= \int 1 dx - \int \sec^2 x dx \\ &= x - \tan x + c \end{aligned}$$

(xiv) $\int \tan^2 x dx$

Solution:

$$\begin{aligned} & \int \tan^2 x dx \\ &= \int (\sec^2 x - 1) dx \\ &= \int \sec^2 x dx - \int dx \\ &= \tan x - x + c \end{aligned}$$